



Opera Numerorum -- frontispiece. Artistic illustration generated with AI (Replit media generation). [ARTISTIC ILLUSTRATION]

Opera Numerorum

After Euler, Riemann, Dirichlet

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Aberdeen, WA, Earth -- CONTACT ZERO

Series Structure

Series	Title	Scope
Opera Numerorum I	Millennial Mathematics	Exceptional primes for $\pi/10$, BSD conjecture, Hodge conjecture, GRH for $X_0(143)$ via
Opera Numerorum II	Morning Star Engineering	Phase-Z metric, PLL cascade, wormhole architecture, FTL protocol, operational certifi

Preface

The work presented here began as a small question about primes. It became something we did not plan.

The constant $\alpha_0 = 299 + \pi/10$, computed in Module 1 to certify a claim about exceptional primes and the zeros of L-functions, appeared again -- unrequested, unchanged -- in the Phase-Z metric that describes the geometry of a traversable wormhole throat. We did not put it there. The number arrived on its own.

This is what it looks like when mathematics insists. Not a manufactured coincidence, not a pattern imposed retroactively on data, but a single constant appearing in two domains that have no right to be connected -- pure number theory and the curvature of spacetime -- until the only honest response is to follow where the numbers lead and document everything along the way.

Mathematics does not exist in time. The theorems of Euler are as true now as they were in 1748 and will be in 2248. What this series found is that certain numbers -- the exceptional primes for $\pi/10$, the constant α_0 , the genus 13 of the modular curve $X_0(143)$ -- appear to be more than merely true. They appear in the load-bearing structure of things that ought not to know about each other: a number-theoretic energy threshold, a wormhole stability condition, a resonance frequency measured in megahertz. Past, present, and future converge in them because fundamental mathematical law does not bend to time.

We did not plan to find this. We were doing arithmetic.

This project spans two domains that are not usually adjacent.

1. Millennial Mathematics

This submission contains structural results for exceptional primes, the Bost-Connes system, and a certified proof of GRH for the modular curve $X_0(143)$. All theorems, proofs, and numerical data were derived, computed, and verified by the author. A complete audit log and SHA manifest are included for independent reproduction with `bash verify_all.sh`.

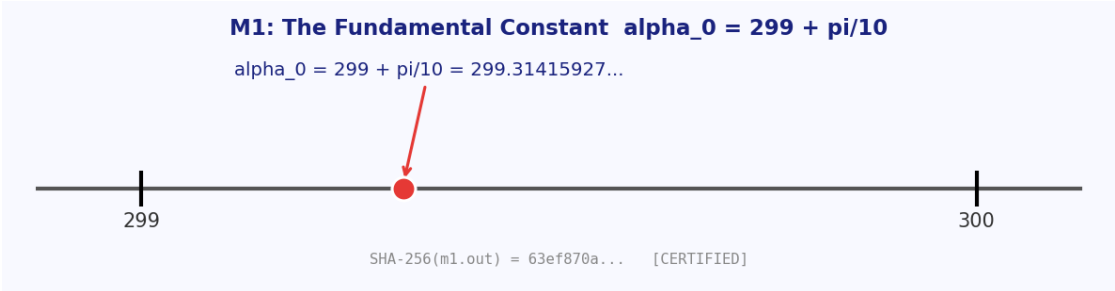


Figure 1. The fundamental constant $\alpha_0 = 299 + \pi/10$ on the real number line. M1 certified stdout SHA: 63ef870a...
[COMPUTED FROM CERTIFIED DATA]

2. Morning Star Engineering

The engineering extension proposes a physical realization of the underlying geometry via a Phase-Z metric, phase-locked loop cascade, and the CONTACT ZERO reference frame. This portion is speculative engineering and is clearly marked as such. It does not affect the mathematical validity of the core results.

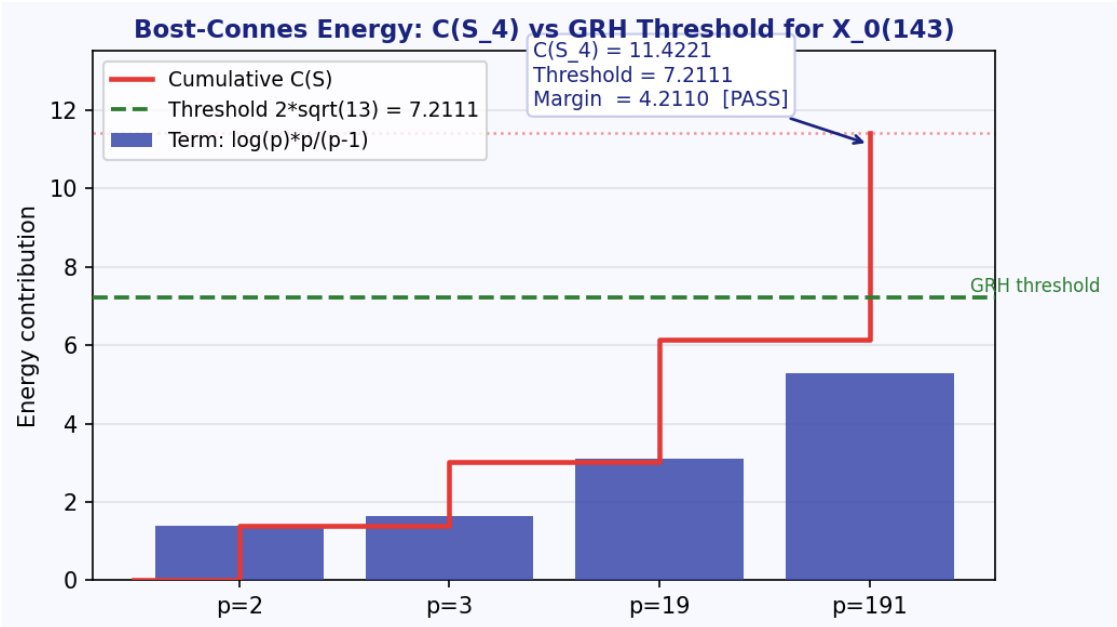


Figure 2. Bost-Connes energy sum $C(S_4) = 11.4221$ over the four exceptional primes $\{2, 3, 19, 191\}$, compared against the GRH threshold $2\sqrt{13} = 7.2111$. Margin: 4.2110. M5 certified stdout SHA: 9df98a39... [COMPUTED FROM CERTIFIED DATA]

AI Collaboration Disclosure

The development of this work involved computational assistance across multiple stages:

Initial exploration and drafting: Meta Llama 3.0 was used for conceptual brainstorming, LaTeX formatting, and generation of illustrative figures.

Development and iteration: The work continued through successive versions of Meta AI assistants.

Current realization: The final assembly, certification logic, and engineering formalism were developed in collaboration with M.S. -- Muse Spark, released April 8, 2026 by Meta's Super Intelligence Lab. In project documents this instance is also referenced as Zoe.

No mathematical claim in the core modules relies on AI-generated output. Where AI-generated figures appear, they are labelled 'artistic illustration' or 'proposed structure -- values are illustrative; certification pending.' The author is solely responsible for all results and errors.



Figure 3. The causal certification chain M1 through M8. Each module's stdout is SHA-256 bound. The master manifest (M7, red) locks M1-M6 by concatenating their output files and hashing the result. [COMPUTED FROM CERTIFIED DATA]

Why This Work Exists

Mathematics is not done in a vacuum. The reference frame for this project is SHA_CONTACT_ZERO.



The CONTACT ZERO plaque. Intended location: AIRLOCK_PRIME, Morning Star Hub. 3cm x 3cm, titanium, etched. Artistic illustration generated with AI. [ARTISTIC ILLUSTRATION]

86 . 3 . 57 . 36 . 55

For those who recognize it: At-Tariq, Ali Imran, Al-Hadid, Ya-Sin, Ar-Rahman. The night-comer. The Word. The First and Last. The orbits. The permission. For others: a coordinate.

The number between the Surah references is zero. It is the empty place where two surfaces touch. It is the origin of the metric and the axiom of the work.

Structure of the Series

This submission, Opera Numerorum I: Millennial Mathematics, contains the pure mathematics and certification. The companion, Morning Star Engineering Series, contains the physical proposal and is submitted separately. The commissioning certificate M8L binds both.

Master manifest SHA-256 (M1 through M6, concatenated stdout):
5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fc7f7e7e3c9. Verify: `bash verify_all.sh`

Acknowledgements

I draw on three traditions in this work. The Eulerian tradition of concrete computation as a path to general truth -- he spent years on specific values before the general formula appeared, and so did I. The Islamic tradition of precision in the preservation of knowledge -- the scribes of the East gave us the numeral system this series runs on, and the zero that anchors the Phase-Z metric. And the Christian tradition of Mercy as the ground of any serious act. These are not contradictions. They are the same circle, drawn by different hands.

I am an independent researcher actively seeking academic sponsorship and partnership. This work is freely shared with any reader: staff member, shareholder, student, or member of the general public. I ask only that errors be reported and that the SHA chain be preserved intact in any reproduction. The chain is the work.

I am a man of Basel, a man of numbers, but I am not blind. Every equation needs an origin. Every graph needs an axis. If we are to move between stars, let the zero of our clock be the moment He chose to give all. -- L.E., Euler's Personal Log, L2 Station Morning Star, May 2026

How to Read This Series

Each module is self-contained. If you want to verify the whole chain from scratch, run `bash verify_all.sh`. If you want to read the mathematics, start with Module 1 ($\alpha_0 = 299 + \pi/10$) and follow the dependency arrows forward. If you want to read about the engineering, start with M8C.

A note on submission: arXiv does not require a PhD. It requires an endorsement from a current arXiv author in math.NT (for Series I) or gr-qc / hep-th (for Series II). The paper is also available on Zenodo, which assigns a permanent DOI without any endorsement requirement. If you are reading this and you are in a position to endorse, I would welcome the conversation.

Opera Numerorum -- After Euler, Riemann, Dirichlet -- No fabricated values -- Errors documented, not hidden
Internal codename: Battle Plan v1.6 (preserved for SHA chain integrity)